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Track assembly for a vehicle

Abstract

A track assembly for a vehicle has a frame; a drive wheel rotatably mounted to the frame; an idler wheel mount pivotally connected to the frame, the idler wheel mount being selectively pivotable between an operating position and a released position; a lock selectively locking the idler wheel mount in the operating position; an idler wheel rotationally connected to the idler wheel mount, the idler wheel being pivotable with the idler wheel mount; and an endless track driven by the drive wheel around the frame. In the operating position of the idler wheel mount, the idler wheel tensions the endless track. Pivoting the idler wheel mount from the operating position to the released position releases at least some tension from the endless track. A vehicle having the track assembly and a method for installing an endless track having derailed from other components of a track assembly are also disclosed.

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Background/Summary

CROSS-REFERENCE (1) The present application claims priority to U.S. Provisional Patent Application No. 62/799,240, filed Jan. 31, 2019, the entirety of which is incorporated herein by reference.

FIELD OF TECHNOLOGY
(1) The present technology relates to track assemblies for vehicles.
BACKGROUND
(2) Side-by-side off-road vehicles (SSVs), all-terrain vehicles (ATVs) and similar vehicles are used for utility and recreational purposes. Some of these vehicles are configured to be interchangeably equipped with ground-engaging wheels or track assemblies, such as to allow a user to equip the vehicle with either option in accordance with terrain conditions and/or desired handling performance Track assemblies are particularly useful for instance when travelling over deep snow as the increased contact area between the track assemblies' tracks and the ground allows for greater floatation.
(3) A track assembly typically includes a frame, a drive wheel rotationally connected to the frame, one or more idler wheels rotationally connected to the frame, an endless track disposed around the frame, the drive wheel and the one or more idler wheels, and a track tensioner. The drive wheel is connected to a wheel hub of the vehicle so as to be driven by the vehicle's motor, and thereby drive the endless track.
(4) The track tensioner is used, as the name suggests, to apply tension to the endless track in order to ensure efficient operation of the track assembly. If there is too much tension on the endless track, the drive wheel, idler wheels, and the shafts and bearings rotationally connecting them to the frame of the track assembly could wear faster. If there is not enough tension on the track, the chance of the endless track derailing from the track assembly during use increase.
(5) Even when the proper amount of tension is applied to the endless track, it is possible that under some operating condition the endless drive track could derail from the track assembly. This could occur for example when the track assembly encounters a sudden deep depression in the ground or when a high side load is applied to the endless track. Such conditions combined with the continued rotation of the endless drive track can lead to the derailment of the endless track. When the endless track derails, the endless track becomes loose and the drive wheel is no longer capable of driving the endless track.
(6) When derailment occurs, the track tensioner which previously assisted in keeping the track from derailing now acts against the user trying to put the endless track back on the track assembly. As such, in order to put the endless track back on, the user has to either force the track back on, which may not be possible, or has to adjust the track tensioner so as to put it in a position that would apply less tension on the endless track.
(7) However, as these track assemblies are provided in off-road vehicles, derailment can occur in an area far from a service station. As such, if the user does not have the proper tools to adjust the track tensioner, putting the endless track back on the track assembly may not be possible. Even if the user has the tools required, the derailment may have occurred in terrain conditions that make the track tensioner difficult to access. Additionally, adjusting the track tensioner can be time

consuming. Also, once the endless track has been put back on the track assembly, the track tensioner has to be adjusted again, this time to put traction back in the endless track in order to avoid another derailment, thus adding to the amount of time required for this operation.

(8) There is therefore a need for a track assembly for a vehicle that facilitates the installation of the endless track back on the track assembly in the event of a derailment of the endless track.

SUMMARY

(9) It is an object of the present technology to ameliorate at least some of the inconveniences present in the prior art.

(10) According to one aspect of the present technology, there is provided a track assembly for a vehicle having a frame; a drive wheel rotatably mounted to the frame about a drive wheel axis; an idler wheel mount pivotally connected to the frame about an idler wheel mount axis, the idler wheel mount axis being vertically below and longitudinally spaced from the drive wheel axis, the idler wheel mount being selectively pivotable between an operating position and a released position about the idler wheel mount axis; and a lock selectively locking the idler wheel mount in the operating position; an idler wheel rotationally connected to the idler wheel mount about an idler wheel axis, the idler wheel being pivotable about the idler wheel mount axis with the idler wheel mount. In the operating position of the idler wheel mount, the idler wheel axis is at a first longitudinal distance and at a first vertical distance from the drive wheel axis. In the released position of the idler wheel mount, the idler wheel axis is at a second longitudinal distance and at a second vertical distance from the drive wheel axis. The first longitudinal distance is greater than the second longitudinal distance. The first vertical distance is greater than the second vertical distance. An endless track is driven by the drive wheel around the frame. In the operating position of the idler wheel mount, the idler wheel tensions the endless track. Pivoting the idler wheel mount from the operating position to the released position releases at least some tension from the endless track.

(11) In some embodiments of the present technology, in the operating position of the idler wheel mount, the idler wheel axis is vertically higher than the idler wheel mount axis.

(12) In some embodiments of the present technology, the lock includes a fastener. The fastener fastens the idler wheel mount to the frame for locking the idler wheel mount in the operating position.

(13) In some embodiments of the present technology, a slide rail is connected to a lower end of the frame. The idler wheel mount is pivotally connected to the slide rail. The idler wheel mount axis extends through the slide rail. The fastener fastens the idler wheel mount to the slide rail for locking the idler wheel mount in the operating position.

(14) In some embodiments of the present technology, a slide rail is connected to a lower end of the frame. The idler wheel mount is pivotally connected to the slide rail. The idler wheel mount axis extends through the slide rail. The lock locks the idler wheel mount to the slide rail for locking the idler wheel mount in the operating position.

(15) In some embodiments of the present technology, the drive wheel is rotatably mounted to an upper end of the frame.

(16) In some embodiments of the present technology, the drive wheel is a drive sprocket.

(17) In some embodiments of the present technology, in the endless track defines a plurality of apertures for receiving teeth of drive sprocket.

(18) In some embodiments of the present technology, an idler wheel position adjuster connects the idler wheel to the idler wheel mount. The idler wheel position adjuster selectively changes a position of the idler wheel on the idler wheel mount for adjusting an amount of tension applied by the idler wheel on the endless track when the idler wheel mount is in the operating position.

(19) In some embodiment of the present technology, when the idler mount is in the operating position the idler wheel position adjuster is adapted for changing a position of the idler wheel between a first position and a second position. The first position of the idler wheel is a position of the idler wheel where the idler wheel is longitudinally furthest from the drive wheel axis. The

second position of the idler wheel is a position of the idler wheel where the idler wheel is longitudinally closest to the drive wheel axis. More tension is released from the endless track by pivoting the idler wheel mount from the operating position to the released position than by moving the idler wheel from the second position to the first position while the idler wheel mount is in the operating position.

(20) In some embodiments of the present technology, the idler wheel is a front idler wheel. The idler wheel axis is a front idler wheel axis. The front idler wheel axis is forward of the drive wheel axis. The idler wheel mount pivots upward and rearward from the operating position to the released position.

(21) In some embodiments of the present technology, in the operating position of the idler wheel mount, the front idler wheel axis is forward of the idler wheel mount axis. In the released position of the idler wheel mount, the front idler wheel axis is rearward of the idler wheel mount axis.

(22) In some embodiments of the present technology, a rear idler wheel is rotationally connected to the frame about a rear idler wheel axis. The rear idler wheel axis is rearward of the drive wheel axis.

(23) In some embodiments of the present technology, the idler wheel is a rear idler wheel. The idler wheel axis is a rear idler wheel axis. The rear idler wheel axis is rearward of the drive wheel axis. The idler wheel mount pivots upward and forward from the operating position to the released position.

(24) In some embodiments of the present technology, in the operating position of the idler wheel mount, the rear idler wheel axis is rearward of the idler wheel mount axis. In the released position of the idler wheel mount, the rear idler wheel axis is forward of the idler wheel mount axis.

(25) In some embodiments of the present technology, a front idler wheel is rotationally connected to the frame about a front idler wheel axis. The front idler wheel axis is forward of the drive wheel axis.

(26) In some embodiments of the present technology, a holder is provided on the idler wheel mount for holding a tool onto the idler wheel mount for applying a torque to the idler wheel mount to pivot the idler wheel mount between the operating and released positions.

(27) In some embodiments of the present technology, the holder is a hole defined in the idler wheel mount.

(28) In some embodiments of the present technology, the idler wheel is a first idler wheel, the idler wheel mount is a first idler wheel mount, and the idler wheel is a first idler wheel. The track assembly also has a second idler wheel mount pivotally connected to the frame about the idler wheel mount axis. The first and second idler wheel mount are selectively pivotable together between the operating position and the released position. A second idler wheel is rotationally connected to the second idler wheel mount about the idler wheel axis. The second idler wheel is pivotable about the idler wheel mount axis with the second idler wheel mount.

(29) In some embodiments of the present technology, the lock is a first lock. The track assembly also has a second lock selectively locking the second idler wheel mount in the operating position.

(30) In some embodiments of the present technology, a first slide rail is connected to a lower end of the frame, and a second slide rail is connected to the lower end of the frame. The first idler wheel mount is pivotally connected to the first slide rail, the second idler wheel mount is pivotally connected to the second slide rail, and the idler wheel mount axis extends through the first and second slide rails.

(31) In some embodiments of the present technology, the idler wheel mount axis is disposed below a line passing through the drive wheel axis and the idler wheel axis.

(32) In some embodiments of the present technology, in the operating position, a projection of the slide rail on a vertically and longitudinally extending plane overlaps a projection of the idler wheel on the plane.

(33) In some embodiments of the present technology, the drive wheel is adapted for connection to a

rotatable axle of a vehicle.
(34) In some embodiments of the present technology, the drive wheel defines a plurality of apertures used for fastening the drive wheel to a wheel hub assembly driven by the rotatable axle of the vehicle.
(35) According to another aspect of the present technology, there is provided a vehicle having a vehicle frame, a motor connected to the vehicle frame, an axle operatively connected to and driven by the motor, a suspension assembly operatively connecting the axle to the vehicle frame, and a track assembly according to one or more of the above embodiments. The drive wheel of the track assembly is connected to and driven by the axle.
(36) According to another aspect of the present technology, there is provided a method for installing an endless track having derailed from other components of a track assembly provided on a vehicle. The track assembly has a frame, a drive wheel rotatably mounted to the frame, and the endless track adapted to be driven by the drive wheel around the frame. The method comprises: unlocking an idler wheel mount from the frame; once the idler wheel mount is unlocked, pivoting the idler wheel mount and an idler wheel rotationally connected to the idler wheel mount about an idler wheel mount axis from an operating position to a released position; once the idler wheel mount is in the released position, realigning the endless track with the other components of the track assembly; once the endless track is realigned, pivoting the idler wheel mount and the idler wheel about the idler wheel mount axis from the released position back to the operating position thereby tensioning the endless track; and once the idler wheel mount is back in the operating position, locking the idler wheel mount in the operating position to the frame.
(37) For purposes of the present application, terms related to spatial orientation when referring to a vehicle and components in relation to the vehicle, such as “forwardly”, “rearwardly”, “left”, “right”, “above” and “below”, are as they would be understood by a driver of the vehicle sitting thereon in a normal driving position, with the vehicle steered straight-ahead.
(38) Embodiments of the present technology each have at least one of the above-mentioned objects and/or aspects, but do not necessarily have all of them. It should be understood that some aspects of the present technology that have resulted from attempting to attain the above-mentioned object may not satisfy this object and/or may satisfy other objects not specifically recited herein.
(39) Additional and/or alternative features, aspects and advantages of embodiments of the present technology will become apparent from the following description, the accompanying drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:
(2) FIG. **1** is a perspective view taken from a front, right side of a side-by-side off-road vehicle having wheels;
(3) FIG. **2** is a left side elevation view of the vehicle of FIG. **1** with the wheels removed;
(4) FIG. **3** is a left side elevation view of the vehicle of FIG. **1** with front and rear track assemblies provided instead of the wheels;
(5) FIG. **4** is a left side elevation view of a rear, right track assembly of the vehicle of FIG. **3** with idler wheel mounts of the track assembly being in an operating position;
(6) FIG. **5** is a right side elevation view of the track assembly of FIG. **4** with the idler wheel mounts in a released position;
(7) FIG. **6** is a left side elevation view of the track assembly of FIG. **4** with the idler wheel mounts

in the operating position and the endless track removed;
(8) FIG. **7** is a perspective view, taken from a rear, right side of the track assembly of FIG. **4** with the idler wheel mounts in the operating position and the endless track removed;
(9) FIG. **8** is a perspective view, taken from a rear, right side of the track assembly of FIG. **4** with the idler wheel mounts in the released position and with a lower portion of the endless track being shown;
(10) FIG. **9** is a top plan view of the track assembly of FIG. **4** with the idler wheel mounts in the operating position and with a lower portion of the endless track being shown in a derailed configuration;
(11) FIG. **10** is a top plan view of the track assembly of FIG. **4** with the idler wheel mounts in the released position and with a lower portion of the endless track being shown in a derailed configuration;
(12) FIG. **11** is a right side elevation view of the track assembly of FIG. **4** illustrating a tool removing a fastener used for locking one of the idler wheel mounts in the operating position;
(13) FIG. **12** is a right side elevation view of the track assembly of FIG. **4** with the fastener removed and illustrating the tool engaging a holder provided in the idler wheel mount for applying a torque to the idler wheel mount to pivot the idler wheel mount to the released position;
(14) FIG. **13** is a right side elevation view of the track assembly of FIG. **4** illustrating the tool having pivoted the idler wheel mounts to the release position;
(15) FIG. **14** is a right side elevation view of a front, left track assembly of the vehicle of FIG. **3** with idler wheel mounts of the track assembly being in an operating position;
(16) FIG. **15** is a left side elevation view of the track assembly of FIG. **14** with the idler wheel mounts in a released position and the endless track removed; and
(17) FIG. **16** is a perspective view taken from a front, left side of the track assembly of FIG. **14** with the idler wheel mounts in a released position and the endless track removed.

DETAILED DESCRIPTION

(18) The present technology will be described with respect to off-road vehicles having four ground-engaging members, two side-by-side seats and a steering wheel (i.e. a side-by-side vehicle (SSV)). However, it is contemplated that at least some aspects of the present technology may apply to other types of vehicles such as, but not limited to, off-road vehicles having a straddle seat and a handle bar (i.e. an all-terrain vehicle (ATV)), off-road vehicles having a single bucket-type seat, off-road vehicles with more than four ground-engaging members, as well as other vehicles that in which the ground-engaging members can be track assemblies.
(19) The general features of an off-road vehicle **40**, specifically a side-by-side vehicle (SSV) **40**, will be described with respect to FIGS. **1** and **2**. The vehicle **40** has a frame **42**. The frame **42** defines a central cockpit area **52** inside which are disposed a driver seat **54** and a passenger seat **56**. In the present implementation, the driver seat **54** is disposed on the left side of the vehicle **40** and the passenger seat **56** is disposed on the right side of the vehicle **40**. However, it is contemplated that the driver seat **54** could be disposed on the right side of the vehicle **40** and that the passenger seat **56** could be disposed on the left side of the vehicle **40**. It is also contemplated that the vehicle **40** could include a single seat for the driver, or a larger number of seats, or a bench accommodating the driver and at least one passenger. The vehicle **40** also includes a roll cage **43** connected to the frame **42** and extending at least partially over the seats **54**, **56**.
(20) The vehicle **40** includes left and right front wheels **44** connected to the frame **42** by a pair of front suspension assemblies **46**. Left and right rear wheels **48** are connected to the frame **42** by a pair of rear suspension assemblies **50**. Each one of the front and rear wheels **44**, **48** has a rim **45** and a tire **47**. The rims **45** and tires **47** of the front wheels **44** may differ in size from the rims and tires of the rear wheels **48**. As will be discussed in more detail below, the front and rear wheels **44**, **48** can be removed, as shown in FIG. **2**, and replaced with front and rear track assemblies **200**, **100** respectively. Ground-engaging members of the type of front and rear track assemblies **200**, **100** are

sometimes referred to as track kits as they are often provided in kits sold separately from the vehicle that the user installs to replace the wheels. However, it is contemplated that the vehicle **40** could be provided with the track assemblies **200**, **100** originally installed at the factory or the vehicle dealership and that these track assemblies **200**, **100** may or may not be able to be replaced by the wheels **44**, **48**. It is also contemplated that the vehicle **40** could have the wheels **44** at the front thereof and the track assemblies **100** at the rear thereof. It is also contemplated that the vehicle **40** could have the track assemblies **100** at the rear thereof, and that the front thereof could be provided with another type of ground engaging members such as skis for example. The front and rear track assemblies **200**, **100** will be described in more detail below.

(21) The vehicle **40** includes a steering wheel **58** operatively connected to the front wheels **44** for controlling an angle of the front wheels **44**. The driver operates the steering wheel **58** from the driver seat **54**. The steering wheel **58** is disposed in front of the driver seat **54**. The vehicle **40** also includes a dashboard **55** disposed forward of the seats **54**, **56**. A throttle operator in the form of a throttle pedal (not shown) is disposed over the floor of the cockpit area **52** below the steering wheel **58** and in front of the driver seat **54**.

(22) As can be seen in FIG. **2**, a motor **62** is connected to the frame **42** in a rear portion of the vehicle **40**. In the present embodiment, the motor **62** is an internal combustion engine but the present technology is not so limited. It is contemplated that the engine **62** could be replaced by a hybrid or electric motor in some implementations. The vehicle **40** includes an engine control module (ECM) for monitoring and controlling various operations of the engine **62**.

(23) As shown in FIG. **2**, the motor **62** is connected to a transmission **64**, specifically a continuously variable transmission (CVT) **64** disposed on a left side of the motor **62**. The CVT **64** is operatively connected to a transaxle **66** to transmit torque from the motor **62** to the transaxle **66**. The transaxle **66** is operatively connected to the front and rear wheels **44**, **48** to propel the vehicle **40**. The transaxle **66** drives a front differential (not shown) The motor **62** and the transmission **64** are supported by the frame **42**. Variants of the vehicle **40** having other transmission types are contemplated.

(24) The transaxle **66** transmits the torque applied thereon to drive the left and right rear wheels **48** via rear axles **35** connected to rear wheel hub assemblies **34**. While the vehicle **40** is described with the rear wheels **48** driving the vehicle **40** when in 2×4 drive mode, it is contemplated that the front wheels **44** could be driven when the vehicle **40** is in 2×4 drive mode in some implementations. Specifically, the transaxle **66** includes left and right half-shafts and a differential connected therebetween for applying torque to the rear driven wheels **48**. The differential is operatively connected between the transmission **64** and the left and right rear driven wheels **48**. Furthermore, in a 4×4 drive mode, the front wheels **44** and the rear wheels **48** are driven. The front wheels **44** are driven via a front differential (not shown) operatively connected to the transaxle **66**. The front differential transmits the torque applied thereon to drive the left and right front wheels **44** via front axles **31** connected to front wheel hub assemblies **30**.

(25) As shown in FIGS. **1**, **2** and **12** each front suspension assembly **46** includes an upper A-arm **24**, a lower A-arm **26**, and a front shock absorber assembly **28**. The upper and lower A-arms **24**, **26** each have one end pivotably connected to the frame **42**. A kingpin **29** (FIG. **2**) is mounted to each opposed ends of the upper and lower A-arms **24**, **26**. The kingpins **29** rotationally support the front wheel hub assemblies **30**. The front wheels **44** are connected to the front wheel hub assemblies **30**.

(26) As shown in FIGS. **2** and **3**, each rear suspension assembly **50** comprises a swing arm **36** and a rear shock absorber assembly **38**. Each swing arm **36** has one end pivotably connected to the frame **42**, about a pivot axis located in front of the rear wheels **48** and extending generally laterally within the frame **42**, and an opposite end rotationally supporting the rear wheel hub assembly **34**. The rear wheels **48** are connected to the rear wheel hub assemblies **34**. The swing arms **36** are connected at mid-length to a torsion bar (not shown) by links.

(27) Turning now to FIGS. **4** to **8**, a rear, right track assembly **100** of the vehicle **40** of FIG. **3** will

be described in more detail. A rear, left track assembly **100** of the vehicle **40** is a mirror image of the rear, right track assembly **100** and as such will not be described herein in detail.
(28) The track assembly **100** has a frame **102**. The frame **102** has a lower, longitudinally extending frame member **104**. As best seen in FIG. **7**, the frame member **104** is connected to a rear cross-member **106** at rear thereof, a central cross-member **108** near a center thereof, and a front cross-member **110** near a front thereof. The front end of the lower frame member **104** is connected to an axle holder **112**. The frame **102** also includes a frame member **114** extending upward and forward from the rear cross-member **106** to an axle holder **116**, a frame member **118** extending upward and rearward from the front cross-member **110** to the axle holder **116**, and a frame member **120** extending upward and slightly forward from the central cross-member **108** to the axle holder **116**.
(29) A drive wheel **122** is rotatably mounted to an upper end of the frame **102** about a drive wheel axis **124**. More specifically, the drive wheel **122** is connected to an axle **126** that is supported by bearings (not shown) disposed inside the axle holder **116**. The drive wheel **122** defines four apertures **128** (FIG. **4**). The drive wheel **122** is disposed on the rear wheel hub assembly **34** such that studs extending from the rear wheel hub assembly **34** are received in the apertures **128**. Nuts are then fastened onto the studs to fasten the drive wheel **122** to the rear wheel hub assembly **34** such that the drive wheel **122** can be driven. In the present implementation, the drive wheel **122** is a drive sprocket **122** having a plurality of radially projecting sprocket teeth **130**. In an alternative embodiment, it is contemplated that in addition to or instead of the sprocket teeth **130**, the drive sprocket **122** could have axially projecting teeth along a periphery thereof on one or both sides thereof.
(30) Two front idler wheels **132** are rotationally connected to the axle holder **112** about a front idler wheel axis **134**. The front idler wheels **132** are disposed on either side of the axle holder **112**. As can be seen in FIG. **4**, the front idler wheel axis **134** is forward and downward of the drive wheel axis **124**.
(31) Two slide rails **136** are connected to the lower end of the frame **102**. As best seen in FIG. **7**, the left slide rail **136** is connected to the left ends of the cross-members **106**, **108**, **110** and the right slide rail **136** is connected to the right ends of the cross-members **106**, **108**, **110**. Two rods **138** are also connected between the two slide rails **136**. The bottom of each slide rail **136** is flat at the rear and center and is upturned at the front.
(32) Two idler wheel mounts **140** are pivotally connected to the rear portions of the slide rails **136** by fasteners **141**. As such, the idler wheel mounts **140** are pivotally connected to the frame **102** via the slide rails **136**. The idler wheel mounts **140** are selectively pivotable together about an idler wheel mount axis **142** between an operating position shown in FIGS. **4**, **6**, **7** and **9** and a released position shown in FIGS. **5**, **8** and **10**. The idler wheel mount axis **142** extends transversely through both slide rails **136** and is disposed vertically below and rearward of the drive wheel axis **124**.
(33) Each idler wheel mount **140** defines a slot **144**. An axle **146** is connected to and extends between the idler wheel mounts **140** via fasteners **148** inserted through the slots **144**. A pair of rear idler wheels **150** is mounted to the axle **146** so as to be rotationally connected to the idler wheel mounts **140**. The rear idler wheels **150** rotate about a rear idler wheel axis **152** defined by the axle **146**. The rear idler wheels **150** and the axle **146** selectively pivot about the idler wheel mount axis **142** together with the idler wheel mounts **140**. As can be seen in FIG. **6**, when the idler wheel mounts **140** are in the operating position, a projection of the slide rails **136** onto a vertically and longitudinally extending plane (i.e. the plane corresponding to the drawing page for FIG. **6**) overlap a projection of the rear idler wheels **150** on this plane. Two idler wheel position adjusters **154** (FIG. **9**) are connected between the idler wheel mounts **140** and the axle **146**. The idler wheel position adjusters **154** selectively adjust a position of the axle **146**, and therefore the rear idler wheels **150**, on the idler wheel mounts **140**. In the present embodiments, the idler wheel position adjusters **154** include fasteners that when turned move the axle **146** forward or rearward by making the fasteners **148** translate in the slots **144**.

(34) It is contemplated that additional idler wheels could be connected to the slide rails **136** and/or the lower portion of the frame **102** at positions longitudinally between the front and rear idler wheels **132**, **150**. It is also contemplated that the idler wheel mounts **140** could be pivotally connected directly to the frame **102** instead of the slide rails **136**. In an embodiment where the idler wheel mounts **140** are pivotally connected directly to the frame **102**, it is contemplated that the slide rails **136** could be omitted and that additional idler wheels could be connected to the lower portion of the frame **102** at positions longitudinally between the front and rear idler wheels **132**, **150**.
(35) The idler wheel mounts **140** and the rear idler wheels **150** will be described in more detail further below.
(36) The rear track assembly **100** also has an endless track **156**. The endless track **156** is disposed around the frame **102**, the drive wheel **122**, the idler wheels **132**, **150**, and the slide rails **136**. More specifically, the drive wheel **122** and the idler wheels **132**, **150** abut an inner surface of the endless track **156** and roll along the inner surface of the endless track **156** when in operation. The slide rails **136** also abut the inner surface of the endless track **156** which slides relative to the endless track **156** when in operation. In order to keep the endless track **156** aligned with the idler wheels **132**, **150** and the slide rails **136**, the endless track **156** defines six continuous internal bands **158** that define two channels **160** and two channels **162** therebetween (see FIG. **8**). The slide rails **136** are received in the channels **160**. The idler wheels **132**, **150** are received in the channels **162**. It is contemplated that some of the continuous internal bands **158** could be replaced by rows of internal lugs. The endless track **156** has a plurality of external lugs **164** on an outer side thereof to provide traction.
(37) The endless track **156** defines a row of apertures **166** (FIG. **8**) along a center thereof. The apertures **166** are engaged by the teeth **130** of the drive sprocket **122** as it turns. As a result, the drive track **156** turns around the frame **102**, the drive wheel **122**, the idler wheels **132**, **150**, and the slide rails **136**, which propels the vehicle **40**. It is contemplated that in addition to or instead of the apertures **166**, the endless track **156** could be provided with rows of internal lugs to be engaged by axially projecting teeth of an alternative embodiment of the drive sprocket **122**.
(38) To mount the rear track assemblies **100** to the vehicle **40**, the rear of the vehicle **40** is first raised, with a jack or a lift for example, such that the rear wheels **28** no longer contact the ground. The rear wheels **48** are unfastened from their corresponding rear wheel hub assemblies **34** and removed. The rear track assemblies **100** are then positioned over the rear wheel hub assemblies **34** and the drive sprockets **122** are fastened to their corresponding rear wheel hub assemblies **34** by studs of the rear hull assemblies **34** inserted through the apertures **128** in the drive sprockets **122** and by nuts as described above. A rotation limiting device (also commonly referred to as an “anti-rotation device”, not shown) is connected between the frame **102** of each rear track assemblies **100** and the frame **42** of the vehicle **40**. The rotation limiting devices limit the rotation of the rear track assemblies **100** about the drive wheel axes **126**.
(39) Returning to FIGS. **4** to **10**, the idler wheel mounts **140** and the rear idler wheels **150** of the rear, right track assembly **100** will now be described in more detail.
(40) As described above, the idler wheel mounts **140** are pivotable, with the rear idler wheels **150**, about the idler wheel mount axis **142** between the operating position shown in FIGS. **4**, **6**, **7** and **9** and the released position shown in FIGS. **5**, **8** and **10**. In the operating position of the idler wheel mounts **140**, the rear idler wheels **150** tension the endless track **156**. The amount of tension applied by the rear idler wheels **150** in the operating position of the idler wheel mounts **140** can be adjusted by the idler wheel position adjusters **154**. By moving the rear idler wheels **150** further rearward using the idler wheel position adjusters **154**, the tension in the endless track **156** is increased. By moving the rear idler wheels **150** further forward using the idler wheel position adjusters **154**, the tension in the endless track **156** is decreased. The idler wheel position adjusters **154** allow for small incremental adjustments of the tension in the endless track **156**. By pivoting the idler wheel mounts

140 from the operating position to the released position, at least some of the tension in the endless track **156** is released. In the present embodiment, the amount of tension release in the endless track **156** is sufficient to allow quick and easy removal of the endless track **156** from the rest of the rear track assembly **100** and also quick and easy installation of the endless track **156** onto the rest of the rear track assembly **100**. As will be described in greater detail further below, moving the idler wheel mounts **140** to the released position also permits realignment of the endless track **100** with the other components of the rear track assembly **100** should the endless track **156** have derailed. In one embodiment, the rear idler wheels **150** do not tension the endless track **156** when the idler wheel mounts **140** are in the released position. Pivoting the idler wheel mounts **140** from the operating position to the released position is a faster and easier way of releasing the tension in the endless track **156** than by moving the rear idler wheels **150** forward using the idler wheel position adjusters **154**. In the present embodiment, more tension can be released from the endless track **156** by pivoting the idler wheel mounts **140** from the operating position to the released position than can be achieved by moving the rear idler wheels **150** to their forwardmost position using the idler wheel position adjusters **154**.

(41) As best seen in FIG. **5**, the idler wheel mounts **140** have a generally obtuse-L shape (i.e. hockey-stick shape). When the idler wheel mounts **140** are in the operating position, the rear idler wheel axis **152** is rearward of the drive wheel axis **124** and the idler wheel mount axis **142**. When the idler wheel mounts **140** are in the released position, the rear idler wheel axis **152** is still rearward of the drive wheel axis **124** but is forward of the idler wheel mount axis **142**. In both the operating and released positions, the rear idler wheel axis **152** is vertically higher than the idler wheel mount axis **142**. As can be seen, from the operating position, the idler wheel mounts **140** pivot upward and forward about the idler wheel mount axis **142** to the released position. As such, the vertical and longitudinal distances between the rear idler wheel axis **152** and the drive wheel axis **124** are greater when the idler wheel mounts **140** are in the operating position than when the idler wheel mounts **140** are in the released position. As can be seen in FIGS. **5** and **6**, in both the operating (FIG. **6**) and released (FIG. **5**) positions of the idler wheel mounts **140**, the idler wheel mount axis **142** is disposed below a line **168** passing through the drive wheel axis **124** and the rear idler wheel axis **152**. In the present embodiment, the idler wheel mounts **140** pivot about 80 degrees from the operating position to the released position. In other embodiments it is contemplated that the idler wheel mounts **140** pivot by an angle of more than 70 degrees from the operating position to the released position. Other angles are also contemplated. In some embodiments, the idler wheel mounts **140** pivot from the operating position to the released position by an angle sufficient to allow the endless track **156** to be manually pulled back in position after having derailed.

(42) In order to lock the idler wheel mounts **140** in the operating position, locks **170** are provided to lock the idler wheel mounts **140** to the slide rails **136**. In the present embodiment, the locks **170** include fasteners **172**. For each idler wheel mount **140**, the fastener **172** is inserted through an aperture **173** (FIG. **8**) in the idler wheel mount **140** and a corresponding aperture **174** (FIG. **5**) in the slide rail **136** thereby fastening the idler wheel mount **140** to the slide rail **136** to lock the idler wheel mount **140** in the operating position. As such, the fastener **172** fastens the idler wheel mount **140** to the frame **102** of the rear track assembly **100** via the slide rail **136** to lock the idler wheel mount **140** in the operating position. In the present embodiment, when the idler wheel mounts **140** are locked in the operating position, the fasteners **172** are located rearward and upward of the idler wheel mount axis **142** and forward and downward of the rear idler wheel axis **152** for most positions of the rear idler wheel **150**. Other types of locks are contemplated, such as latches or quick release pins for example. It is also contemplated that only one of the idler wheel mounts **140** could be provided with a lock **170**.

(43) In order to assist the user in applying the torque necessary to move the idler wheel mounts **140** between the operating position to the released position, the idler wheel mounts **140** are each

provided with a holder **176**. Each holder **176** is designed to hold a tool onto the corresponding idler wheel mount **140** such that the necessary torque can be applied. In the present embodiment, the holders **176** are square holes **176** that are sized to receive the drive shank of a ratchet wrench **178** (FIG. **11**). The square holes **176** are disposed near the lower, rear corners of the idler wheel mounts **140** disposed in the operating position. Although the square holes **176** are sized to receive a drive shank of a ratchet wrench **178**, almost any tool or device that can be inserted in the square holes **176** to apply a torque on the idler wheel mounts **140** can be used. It is contemplated that the holders **176** could be something other than a hole. For example, the holders **176** could be a pin or another type of projection that can be grabbed by pliers or a wrench used to unfasten fasteners **172**. It is contemplated that only one of the idler wheel mounts **140** could be provided with a holder **176**. It is also contemplated that in some embodiments, the holders **176** could be omitted.

(44) As discussed in the background section, under certain operating conditions it is possible for the endless track to derail from the other components of the track assembly. FIGS. **9** and **10** show the endless track **156** of the rear track assembly **100** having derailed. As can be seen, the slide rails **136** and the idler wheels **132**, **150** are no longer received in their corresponding channels **160**, **162**. When this happens for the rear track assembly **100**, the endless track **156** can be installed again by first pivoting the idler wheel mounts **140** from the operating position to the released position as will now be described with reference to FIGS. **11** to **13**. Although the method for installing the endless track **156** having derailed from the other components of the rear track assembly **100** provided on the vehicle **40** will be described using the ratchet wrench **178**, it is contemplated that the method could be performed with a different tool or device, with multiple tools or devices and, in some embodiments, without any tools.

(45) When the endless track **156** derails, the first thing to do is to stop the engine of the vehicle **40** and ensure that the vehicle is stable. Then, the idler wheel mounts **140** are unlocked from the frame **102** of the rear track assembly **100**. This is done by removing the fasteners **172** connecting the idler wheel mounts **140** to the slide rails **136** using the ratchet wrench **178** which has been provided with the proper socket as seen in FIG. **11**. Once the idler wheel mounts **140** are unlocked, the idler wheel mounts **140** and the rear idler wheels **150** are pivoted about the idler wheel mount axis **142** from the operating position to the released position. This movement is helped by inserting the drive shank of the ratchet wrench **178** into the square hole (i.e. holder **176**) of one of the idler wheel mounts **140** as shown in FIG. **12**, and then applying torque to the idler wheel mounts **140** using the ratchet wrench **178** to pivot the idler wheel mounts **140**, and the rear idler wheels **150**, upward and forward to the released position as shown in FIG. **13**. Once the idler wheel mounts **140** are in the released position, the ratchet wrench **178** is remove from the holder **176** and the endless track **156** is realigned with the other components of the rear track assembly **100**. More specifically, the slide rails **136** and the idler wheels **132**, **150** are placed in their corresponding channels **160**, **162** and teeth **130** of the drive wheel **122** are place inside apertures **166**. Once the endless track **156** is realigned, the ratchet wrench **178** is reengaged with the holder **176**, then the idler wheel mounts **140** and the rear idler wheels **150** are pivoted about the idler wheel mount axis **142** from the released position back to the operating position using the ratchet wrench **178**. As a result, the endless track is tensioned. Finally, the idler wheel mounts **140** are locked in the operating position to the frame **102** of the track assembly. This is done by fastening the idler wheel mounts **140** to the slide rails **136** using the fasteners **172** and the ratchet wrench **178** which has been provided with the proper socket.

(46) Turning now to FIGS. **14** to **16**, a front, left track assembly **200** of the vehicle **40** of FIG. **3** will be described in more detail. A front, right track assembly **200** of the vehicle **40** is a mirror image of the front, left track assembly **200** and as such will not be described herein in detail. As can be seen by comparing the front right track assembly **200** to the rear right track assembly **100**, the front track assembly **200** is shorter in length than the rear track assembly **100**. Also, the lower front and rear portions of the front track assembly **200** extend diagonally upward from a central flat portion. Both

of these features (i.e. shorter length and diagonally upward lower portions) facilitate steering of the front track assembly **200**.

(47) The track assembly **200** has a frame **202**. The frame **202** has a lower, longitudinally extending frame member **204**. As best seen in FIG. **16**, the frame member **204** is connected to a front cross-member **206** at front thereof and a rear cross-member **210** near a rear thereof. The rear end of the lower frame member **204** is connected to an axle holder **212**. The frame **202** also includes a frame member **214** extending upward and rearward from a front of the lower frame member **204** to an axle holder **216** and a frame member **218** extending upward and forward from a rear of the lower frame member **204** to the axle holder **216**.

(48) A drive wheel **222** is rotatably mounted to an upper end of the frame **202** about a drive wheel axis **224**. More specifically, the drive wheel **222** is connected to an axle **226** that is supported by bearings (not shown) disposed inside the axle holder **216**. The drive wheel **222** defines four apertures **228**. The drive wheel **222** is disposed on the front wheel hub assembly **30** such that studs extending from the front wheel hub assembly **30** are received in the apertures **228**. Nuts are then fastened onto the studs to fasten the drive wheel **222** to the front wheel hub assembly **30** such that the drive wheel **222** can be driven. In the present implementation, the drive wheel **222** is a drive sprocket **222** having a plurality of radially projecting sprocket teeth **230**. In an alternative embodiment, it is contemplated that in addition to or instead of the sprocket teeth **230**, the drive sprocket **222** could have axially projecting teeth along a periphery thereof on one or both sides thereof.

(49) Two rear idler wheels **232** are rotationally connected to the axle holder **212** about a rear idler wheel axis **234**. The rear idler wheels **232** are disposed on either side of the axle holder **212**. As can be seen in FIG. **15**, the rear idler wheel axis **234** is rearward and downward of the drive wheel axis **224**.

(50) Two slide rails **236** are connected to the lower end of the frame **202**. As best seen in FIG. **16**, the left slide rail **236** is connected to the left ends of the cross-members **206**, **210** and the right slide rail **236** is connected to the right ends of the cross-members **206**, **210**. Two axles **238** are also connected between the two slide rails **236** and longitudinally between the cross-members **206**, **210**. Two idler wheels **239** are mounted on the rear axle **238** and one idler wheel **239** is mounted on the front axle **238**. The bottom of each slide rail **236** is flat at the center, is upturned at the rear, and has a front portion that extends upward and forward from the center portion.

(51) Two idler wheel mounts **240** are pivotally connected to the front portions of the slide rails **236** by fasteners **241**. As such, the idler wheel mounts **240** are pivotally connected to the frame **202** via the slide rails **236**. The idler wheel mounts **240** are selectively pivotable together about an idler wheel mount axis **242** between an operating position shown in FIG. **14** and a released position shown in FIGS. **15** and **16**. The idler wheel mount axis **242** extends transversely through both slide rails **236** and is disposed vertically below and forward of the drive wheel axis **224**.

(52) Each idler wheel mount **240** defines a slot **244**. An axle **246** is connected to and extends between the idler wheel mounts **240** via fasteners **248** inserted through the slots **244**. A pair of front idler wheels **250** is mounted to the axle **246** so as to be rotationally connected to the idler wheel mounts **240**. The front idler wheels **250** rotate about a front idler wheel axis **252** defined by the axle **246**. The front idler wheels **250** and the axle **246** selectively pivot about the idler wheel mount axis **242** together with the idler wheel mounts **240**. As can be seen in FIG. **14**, when the idler wheel mounts **240** are in the operating position, a projection of the slide rails **236** onto a vertically and longitudinally extending plane (i.e. the plane corresponding to the drawing page for FIG. **14**) overlap a projection of the front idler wheels **250** on this plane. Two idler wheel position adjusters **254** are connected between the idler wheel mounts **240** and the axle **246**. The idler wheel position adjusters **254** selectively adjust a position of the axle **246**, and therefore the front idler wheels **250**, on the idler wheel mounts **240**. In the present embodiments, the idler wheel position adjusters **254** include fasteners that when turned move the axle **246** forward or rearward by making the fasteners

248 translate in the slots **244**.
(53) It is contemplated that more or less idler wheels **239** could be provided. It is also contemplated that the idler wheels **239** could be omitted. It is also contemplated that the idler wheels **239** could be connected to the lower frame member **204** instead of the slide rails **236** via the axles **236**. It is also contemplated that the idler wheel mounts **240** could be pivotally connected directly to the frame **202** instead of the slide rails **236**. In an embodiment where the idler wheel mounts **240** are pivotally connected directly to the frame **202**, it is contemplated that the slide rails **136** could be omitted and that the idler wheels **239** would be connected to the lower portion of the frame **202**.
(54) The idler wheel mounts **240** and the rear idler wheels **250** will be described in more detail further below.
(55) The front track assembly **200** also has an endless track **256**. The endless track **256** is disposed around the frame **202**, the drive wheel **222**, the idler wheels **232**, **239**, **250**, and the slide rails **236**. More specifically, the drive wheel **222** and the idler wheels **232**, **239**, **250** abut an inner surface of the endless track **256** and roll along the inner surface of the endless track **256** when in operation. The slide rails **236** also abut the inner surface of the endless track **256** which slides relative to the endless track **256** when in operation. In order to keep the endless track **256** aligned with the idler wheels **232**, **239**, **250** and the slide rails **236**, the endless track **256** defines six continuous internal bands that define four channels therebetween (now shown but similar to those of the endless track **156**). The slide rails **236** are received in two of the channels. The idler wheels **232**, **239**, **250** are received in the other two channels. It is contemplated that some of the continuous internal bands could be replaced by rows of internal lugs. The endless track **256** has a plurality of external lugs **264** on an outer side thereof to provide traction.
(56) The endless track **256** defines a row of apertures (not shown but similar to the apertures **166** of the endless track **156**) along a center thereof. The apertures are engaged by the teeth **230** of the drive sprocket **122** as it turns. As a result, the drive track **256** turns around the frame **202**, the drive wheel **222**, the idler wheels **232**, **239**, **250**, and the slide rails **236**, which propels the vehicle **40**. It is contemplated that in addition to or instead of these apertures, the endless track **156** could be provided with rows of internal lugs to be engaged by axially projecting teeth of an alternative embodiment of the drive sprocket **222**.
(57) The front track assemblies **200** are mounted to the vehicle **40** by fastening the drive sprockets **222** to the front wheel hub assemblies **30**. This installation is similar to the manner in which the rear track assemblies **100** are mounted to the vehicle **40** and as such will not be described herein.
(58) The idler wheel mounts **240** and the front idler wheels **250** of the front, left track assembly **200** will now be described in more detail.
(59) As described above, the idler wheel mounts **240** are pivotable, with the front idler wheels **250**, about the idler wheel mount axis **242** between the operating position shown in FIG. **14** and the released position shown in FIGS. **15** and **16**. In the operating position of the idler wheel mounts **240**, the front idler wheels **250** tension the endless track **256**. The amount of tension applied by the front idler wheels **250** in the operating position of the idler wheel mounts **240** can be adjusted by the idler wheel position adjusters **254**. By moving the front idler wheels **250** further forward using the idler wheel position adjusters **254**, the tension in the endless track **256** is increased. By moving the front idler wheels **250** further rearward using the idler wheel position adjusters **254**, the tension in the endless track **256** is decreased. The idler wheel position adjusters **254** allow for small incremental adjustments of the tension in the endless track **256**. By pivoting the idler wheel mounts **240** from the operating position to the released position, at least some of the tension in the endless track **256** is released. In the present embodiment, the amount of tension release in the endless track **256** is sufficient to allow quick and easy removal of the endless track **256** from the rest of the front track assembly **200** and also quick and easy installation of the endless track **256** onto the rest of the front track assembly **200**. As will be described in greater detail further below, moving the idler wheel mounts **240** to the released position also permits realignment of the endless track **200** with

the other components of the front track assembly **200** should the endless track **256** have derailed. In one embodiment, the front idler wheels **250** do not tension the endless track **256** when the idler wheel mounts **240** are in the released position. Pivoting the idler wheel mounts **240** from the operating position to the released position is a faster and easier way of releasing the tension in the endless track **256** than by moving the front idler wheels **250** rearward using the idler wheel position adjusters **254**. In the present embodiment, more tension can be released from the endless track **256** by pivoting the idler wheel mounts **240** from the operating position to the released position than can be achieved by moving the front idler wheels **250** to their rearmost position using the idler wheel position adjusters **254**.

(60) As best seen in FIG. **15**, the idler wheel mounts **240** are generally straight. When the idler wheel mounts **240** are in the operating position, the front idler wheel axis **252** is forward of the drive wheel axis **224** and the idler wheel mount axis **242**. When the idler wheel mounts **240** are in the released position, the front idler wheel axis **252** is still forward of the drive wheel axis **224** but is rearward of the idler wheel mount axis **242**. In both the operating and released positions, the front idler wheel axis **252** is vertically higher than the idler wheel mount axis **242**. As can be seen, from the operating position, the idler wheel mounts **240** pivot upward and rearward about the idler wheel mount axis **242** to the released position. As such, the vertical and longitudinal distances between the front idler wheel axis **252** and the drive wheel axis **224** are greater when the idler wheel mounts **240** are in the operating position than when the idler wheel mounts **240** are in the released position. As can be seen in FIGS. **14** and **15**, in both the operating (FIG. **14**) and released (FIG. **15**) positions of the idler wheel mounts **240**, the idler wheel mount axis **242** is disposed below a line **268** passing through the drive wheel axis **224** and the front idler wheel axis **252**. In the present embodiment, the idler wheel mounts **240** pivot about 60 degrees from the operating position to the released position. In other embodiments it is contemplated that the idler wheel mounts **240** pivot by an angle of more than 50 degrees from the operating position to the released position. Other angles are also contemplated. In some embodiments, the idler wheel mounts **240** pivot from the operating position to the released position by an angle sufficient to allow the endless track **256** to be manually pulled back in position after having derailed.

(61) In order to lock the idler wheel mounts **240** in the operating position, locks **270** are provided to lock the idler wheel mounts **240** to the slide rails **236**. In the present embodiment, the locks **270** include fasteners **272**. For each idler wheel mount **240**, the fastener **272** is inserted through an aperture **273** (FIG. **15**) in the idler wheel mount **240** and a corresponding aperture **274** (FIG. **15**) in the slide rail **236** thereby fastening the idler wheel mount **240** to the slide rail **236** to lock the idler wheel mount **240** in the operating position. As such, the fastener **272** fastens the idler wheel mount **240** to the frame **202** of the front track assembly **200** via the slide rail **236** to lock the idler wheel mount **240** in the operating position. In the present embodiment, when the idler wheel mounts **240** are locked in the operating position, the fasteners **272** are located forward and upward of the idler wheel mount axis **242** and rearward and downward of the front idler wheel axis **252**. Other types of locks are contemplated, such as latches or quick release pins for example. It is also contemplated that only one of the idler wheel mounts **240** could be provided with a lock **270**.

(62) In order to assist the user in applying the torque necessary to move the idler wheel mounts **240** between the operating position to the released position, the left idler wheel mount **240** is provided with a holder **276** (FIG. **15**). The holder **276** is designed to hold a tool onto the left idler wheel mount **240** such that the necessary torque can be applied. In the present embodiment, the holder **276** is a square hole **276** that is the same size as the square holes **176** of the idler wheel mounts **140** described above in order to receive the drive shank of the ratchet wrench **178**. The square hole **276** are disposed near the center of the left idler wheel mount **240**. It is contemplated that the holder **276** could be something other than a hole. For example, the holder **276** could be a pin or another type of projection that can be grabbed by pliers or a wrench used to unfasten fasteners **172**. It is contemplated that both idler wheel mounts **140** could be provided with a holder **276**. It is also

contemplated that in some embodiments, the holder **276** could be omitted.
(63) When the endless track **256** derails from the front track assembly **200**, the drive track **256** can be installed again by pivoting the idler wheel mounts **240** about the idler wheel mount axis **242** upward and rearward from the operating position to the released position. As the method for installing the endless track **256** having derailed from the other components of the front track assembly **200** provided on the vehicle **40** is similar to the method for installing the endless track **156** having derailed from the other components of the rear track assembly **100** described above, this method will not be described in detail herein.
(64) Modifications and improvements to the above-described embodiments of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting. The scope of the present technology is therefore intended to be limited solely by the scope of the appended claims.

Claims

1. A track assembly for a vehicle comprising: a frame; a drive wheel rotatably mounted to the frame about a drive wheel axis; an idler wheel mount pivotally connected to the frame about an idler wheel mount axis, the idler wheel mount axis being vertically below and longitudinally spaced from the drive wheel axis, the idler wheel mount being selectively pivotable between an operating position and a released position about the idler wheel mount axis, the idler wheel mounting pivoting upward and rearward from the operating position to the released position; a lock selectively locking the idler wheel mount in the operating position; a front idler wheel rotationally connected to the idler wheel mount about a front idler wheel axis, the front idler wheel axis being forward of the drive wheel axis, the front idler wheel being pivotable about the idler wheel mount axis with the idler wheel mount, in the operating position of the idler wheel mount, the front idler wheel axis being forward of the idler wheel mount axis, and the front idler wheel axis being at a first longitudinal distance and at a first vertical distance from the drive wheel axis, in the released position of the idler wheel mount, the front idler wheel axis being rearward of the idler wheel mount axis, and the front idler wheel axis being at a second longitudinal distance and at a second vertical distance from the drive wheel axis, the first longitudinal distance being greater than the second longitudinal distance, the first vertical distance being greater than the second vertical distance; and an endless track driven by the drive wheel around the frame, in the operating position of the idler wheel mount, the front idler wheel tensioning the endless track, and pivoting the idler wheel mount from the operating position to the released position releasing at least some tension from the endless track.
2. The track assembly of claim 1, wherein, in the operating position of the idler wheel mount, the front idler wheel axis is vertically higher than the idler wheel mount axis.
3. The track assembly of claim 1, wherein: the lock includes a fastener; and the fastener fastening the idler wheel mount to the frame for locking the idler wheel mount in the operating position.
4. The track assembly of claim 3, further comprising a slide rail connected to a lower end of the frame; wherein: the idler wheel mount is pivotally connected to the slide rail; the idler wheel mount axis extends through the slide rail; and the fastener fastens the idler wheel mount to the slide rail for locking the idler wheel mount in the operating position.
5. The track assembly of claim 4, wherein, in the operating position, a projection of the slide rail on a vertically and longitudinally extending plane overlaps a projection of the front idler wheel on the plane.
6. The track assembly of claim 1, further comprising a slide rail connected to a lower end of the frame; wherein: the idler wheel mount is pivotally connected to the slide rail; the idler wheel mount axis extends through the slide rail; and the lock locks the idler wheel mount to the slide rail for locking the idler wheel mount in the operating position.

7. The track assembly of claim 1, wherein the drive wheel is rotatably mounted to an upper end of the frame.
8. The track assembly of claim 1, further comprising an idler wheel position adjuster connecting the front idler wheel to the idler wheel mount, the idler wheel position adjuster selectively changing a position of the front idler wheel on the idler wheel mount for adjusting an amount of tension applied by the front idler wheel on the endless track when the idler wheel mount is in the operating position.
9. The track assembly of claim 8, wherein, in response to the idler wheel mount being in the operating position: the idler wheel position adjuster is adapted for changing a position of the front idler wheel between a first position and a second position, the first position of the front idler wheel being a position of the front idler wheel where the front idler wheel is longitudinally furthest from the drive wheel axis, and the second position of the front idler wheel being a position of the front idler wheel where the front idler wheel is longitudinally closest to the drive wheel axis; and wherein more tension is released from the endless track by pivoting the idler wheel mount from the operating position to the released position than by moving the front idler wheel from the second position to the first position while the idler wheel mount is in the operating position.
10. The track assembly of claim 8, wherein the idler wheel mount is selectively pivotable between the operating position and the released position about the idler wheel mount axis with the idler wheel position adjuster keeping the front idler wheel in a fixed position on the idler wheel mount.
11. The track assembly of claim 1, further comprising a holder provided on the idler wheel mount for holding a tool onto the idler wheel mount for applying a torque to the idler wheel mount to pivot the idler wheel mount between the operating and released positions.
12. The track assembly of claim 1, wherein the idler wheel mount axis is disposed below a line passing through the drive wheel axis and the front idler wheel axis.
13. The track assembly of claim 1, wherein the drive wheel is adapted for connection to a rotatable axle of a vehicle.
14. The track assembly of claim 13, wherein the drive wheel defines a plurality of apertures used for fastening the drive wheel to a wheel hub assembly driven by the rotatable axle of the vehicle.
15. The track assembly of claim 1, wherein: the frame defines a first aperture; the idler wheel mount defines a second aperture; in the operating position, the second aperture is aligned with the first aperture; and in the released position, the second aperture is not aligned with the first aperture.
16. A method for installing an endless track having derailed from other components of a track assembly provided on a vehicle, the track assembly having: a frame; a drive wheel rotatably mounted to the frame about a drive wheel axis; and the endless track adapted to be driven by the drive wheel around the frame; the method comprising: unlocking an idler wheel mount from the frame; once the idler wheel mount is unlocked, pivoting the idler wheel mount and a front idler wheel upward and rearward about an idler wheel mount axis from an operating position to a released position, the front idler wheel being rotationally connected to the idler wheel mount about a front idler wheel axis, the front idler wheel axis being forward of the drive wheel axis; once the idler wheel mount is in the released position and with the idler wheel mount in the released position, realigning the endless track with the other components of the track assembly, in the released position, the front idler wheel axis being rearward of the idler wheel mount axis; once the endless track is realigned, pivoting the idler wheel mount and the front idler wheel about the idler wheel mount axis from the released position back to the operating position thereby tensioning the endless track, in the operating position, the front idler wheel axis being forward of the idler wheel mount axis; and once the idler wheel mount is back in the operating position, locking the idler wheel mount in the operating position to the frame.
17. The method of claim 16, wherein pivoting the idler wheel mount and the front idler wheel about the idler wheel mount axis from the released position back to the operating position comprises engaging the idler wheel mount with a tool and applying torque to the idler wheel mount

using the tool.

18. A track assembly for a vehicle comprising: a frame; a slide rail connected to a lower end of the frame; a drive wheel rotatably mounted to the frame about a drive wheel axis; an idler wheel mount pivotally connected to the slide rail about an idler wheel mount axis, the idler wheel mount axis extending through the slide rail, the idler wheel mount axis being vertically below and longitudinally spaced from the drive wheel axis, the idler wheel mount defining an aperture, the idler wheel mount being selectively pivotable between an operating position and a released position about the idler wheel mount axis, in the operating position, the aperture being longitudinally spaced from and forward of the idler wheel mount axis; a lock selectively locking the idler wheel mount in the operating position, the lock including a fastener, the fastener being inserted through the aperture for fastening the idler wheel mount to the frame for locking the idler wheel mount in the operating position; an idler wheel rotationally connected to the idler wheel mount about an idler wheel axis, the front idler wheel being pivotable about the idler wheel mount axis with the idler wheel mount, in the operating position of the idler wheel mount, the idler wheel axis being at a first longitudinal distance and at a first vertical distance from the drive wheel axis, in the released position of the idler wheel mount, the idler wheel axis being at a second longitudinal distance and at a second vertical distance from the drive wheel axis, the first longitudinal distance being greater than the second longitudinal distance, the first vertical distance being greater than the second vertical distance; and an endless track driven by the drive wheel around the frame, in the operating position of the idler wheel mount, the idler wheel tensioning the endless track, and pivoting the idler wheel mount from the operating position to the released position releasing at least some tension from the endless track.

19. The track assembly of claim 18, wherein: the aperture is a first aperture; the slide rail defines a second aperture; in the operating position, the second aperture is aligned with the first aperture; in the released position, the second aperture is not aligned with the first aperture; and the fastener is inserted through the first aperture and the second aperture for fastening the idler wheel mount to the slide rail for locking the idler wheel mount in the operating position.

20. The track assembly of claim 18, wherein, in the operating position of the idler wheel mount, the idler wheel axis is vertically higher than the idler wheel mount axis.